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Kinetics modelling of a self-sustained discharge KrF laser

D S Ong†, T Y Tou and K S Low

Institute of Advanced Studies, University of Malaya, 50603 Kuala Lumpur, Malaysia

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Abstract. A kinetics model was improved to describe the operation of a self-sustained discharge KrF laser in a Kr–F_2–He gas mixture. The improvements include considerations of the effects of the discharge excited-state manifolds of Kr atoms and F_2 molecules on the electron energy distribution function, the KrF excimer formation and electron-impact ionization of the doubly excited $Kr^{**}(5p)$ species, and a two-state model for the upper laser level. The model predictions were compared to the experimental results from a UV-pre-ionized, transversely excited KrF laser, and reasonably good agreements were obtained between the predicted and experimental laser output energy and temporal pulse characteristics.

1. Introduction

Since the first demonstration of the KrF laser [1], kinetics modelling [2–9] has been performed actively to investigate the various factors which determine its output efficiency and characteristics under different operating conditions. Over the years, important kinetics such as the vibrational relaxation [10–13], collisional mixing of B and C states [14, 15], electron-collision quenching [16, 17], depletion of F_2 [18] and electron photodetachment [19] have been incorporated into these models. Although these models have become a useful tool for kinetics simulation, they were mainly applied to the electron-beam-sustained discharge KrF laser. In comparison, the existing models [20, 21] predicted at least twice as much output energy and a much larger pulse duration than the experimental results. The next model [22] assumed the self-sustained discharge to be in a steady state so that the main kinetics processes could be described by a set of time-independent algebraic equations. In this model, the general trend of the laser output energy was studied but the predicted and measured results were not compared in terms of absolute values. Thus, it is difficult to determine the accuracy of predictions from this steady-state model. The limited modelling effort for a self-sustained discharge KrF laser contrasts with that for the XeCl laser [23–25], for which much better agreement has been obtained between the model and the experiment. In particular, the model of Turner and Smith [23] could predict the laser output energy with an accuracy of better than within 5%.

Although similar discharge-excited kinetics may occur both in the electron-beam-sustained and in the self-sustained discharge KrF lasers, in general, they can differ owing to their operation conditions, such as the buffer gas and discharge behaviour [26]. In view of the need for computer simulation, which provides basic guidelines for designing and optimization of a self-sustained discharge KrF laser, we have embarked on improving these existing models [20–22]. In this effort, besides taking into consideration all the plasma chemistry employed in the early models [21, 22], some relevant kinetics and ideas used in the KrF electron-beam and XeCl discharge models were adopted. For example, the two-state upper laser level [8, 23] and the contribution of the doubly excited $Kr^{**}(5p)$ atoms to the KrF formation and electron density [6] were included in the present model. The discharge formation of highly excited-state manifolds of Kr and F_2 atoms, which has not previously been accounted for, was included in the Boltzmann equation for the calculation of the electron energy distribution function (EEDF).

A UV-pre-ionized, transversely excited KrF laser was designed and characterized, so that the model predictions could be compared with the experimental results. The experimental set-up and measurements are presented in section 2, followed by a brief description of the present KrF model in section 3, the temporal evolution of 16 selected discharge-excited chemical species in section 4, comparisons between the predicted and measured results in section 5, and lastly discussions and a conclusion in section 6.

2. The experiment

Figure 1(*a*) shows the end-on view of the KrF laser assembly. The laser chamber was made out of a 12.5 cm (diameter) cylindrical PVC tube of length 40 cm, the inner wall of which was covered a thin Teflon sheet. The laser channel was formed between a nickel-coated electrode and a stainless-steel wire mesh which were separated by about 1.1 cm and the laser discharge volume was estimated to

† Present address: Department of Electronic & Electrical Engineering, Sheffield University, Mappin Street, Sheffield S1 3JD, UK.

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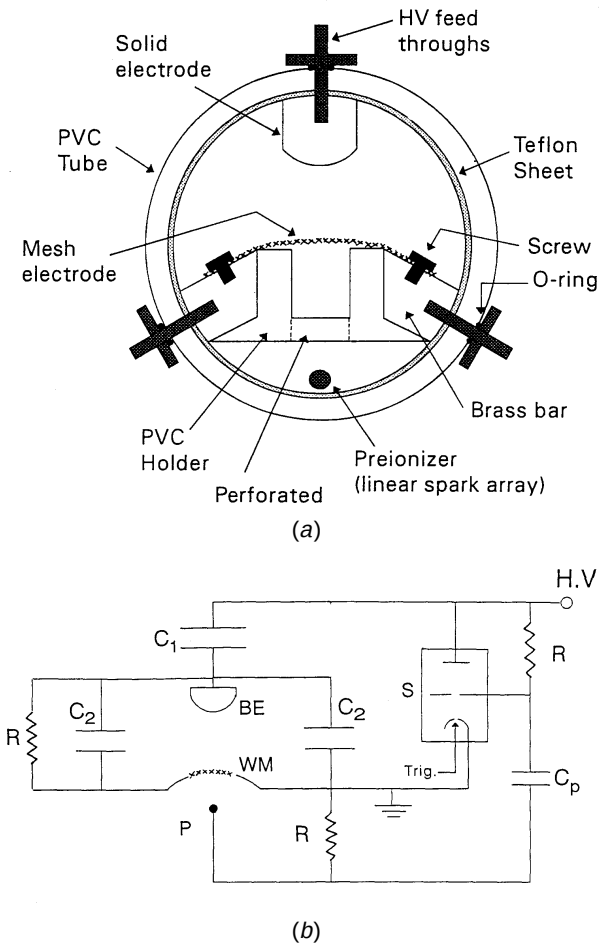


Figure 1. (*a*) An end-on view of the KrF laser chamber, and of the UV-pre-ionizer and electrode arrangements. (*b*) A schematic diagram of the electrical arrangement for the KrF laser: HV, high-voltage charger; R, 5 kΩ resistor; C_1 and C_2, the storage and peaking capacitors of the C–C charge-transfer circuit; C_p the pre-ionizer capacitor; S, the two-stage spark gap [28]; and BE and WM, the solid and wire-mesh laser electrodes.

be $(0.3 \times 27 \times 1.1)$ cm^3. The laser channel was UV-pre-ionized by a linear spark-array placed beneath the wire-mesh electrode. The optical resonator was formed between a quartz window of the laser chamber and a CaF_2-coated back mirror of 100% reflectivity. The laser was operated in a Kr–F_2–He (4.3:0.24:95.46%) gas mixtures at pressures of 1.5–3 bars and the charging voltage varied within the range of 16–24 kV.

The electrical arrangement of this laser is shown in figure 1(*b*), in which a capacitor transfer (C-to-C) circuit was employed for exciting the laser discharge, and C_1 and C_2 were the storage and peaking capacitors respectively. The peaking capacitor C_2 consisted of two arrays of 0.44 nF doorknob capacitors on both sides of the laser chamber. It was found that the capacitance ratio $C_2/C_1 = \frac{1}{3}$ produced the highest laser output. The UV-pre-ionizer was excited by a 2 nF capacitor, C_p. Both the pre-ionizer circuit and the laser circuit were temporally controlled by a two-stage spark gap (S) [27], such that the laser discharge could be delayed by up to 500 ns.

The laser output energy was measured by a calibrated Scientech-390201 calorimeter. It was found that the laser output energy began to decrease after about 30 shots, thus no data would be collected thereafter. The laser chamber was then pumped down and flushed with purified He gas before it was filled with fresh laser gas mixture. The laser output pulse was measured by a fast photodiode (FND100Q, EG&G); the discharge voltage and current by, respectively, a Tektronix P6105 high-voltage probe and a Rogowski coil. These signals were captured by a four-channel Tektronix oscilloscope, and will be presented in section 5 for comparison with the model predictions.

3. The model

This kinetic model assumes spatial uniformity in the laser discharge and employs the bulk of the modelling techniques used by Johnson *et al* [28] for the high-pressure glow discharges. The model can be divided into four sub-models: (i) an electrical circuit coupled with a time-varying plasma resistance; (ii) a steady-state Boltzmann equation for calculation of the electron energy distribution function and electron kinetics rate constants; (iii) a set of kinetic equations governing the formation and loss of various discharge species; and (iv) a photon rate equation for the laser oscillation. A total of 93 kinetic reactions among 30 different discharge species were considered. Table 1 shows all these kinetic reactions, their rate constants and sources or references from which these respective values were obtained for running this model.

As mentioned in section 1, a two-state upper laser level was adopted in this model by which KrF molecules were first formed in high vibrational states as KrF**, which could not be extracted for the laser oscillation. In addition to excited Kr* atoms and Kr$^+$ ions, the doubly excited Kr**(5p) atoms also contributed to the immediate formation of KrF**. The main formation pathways for these highly excited KrF** molecules are the harpooning reactions ($Kr^*/Kr^{**} + F_2$) and three-body ionic recombination ($Kr^+/Kr_2^+ + F^- + He$). Owing to the lack of relevant collision cross section data, the formation of KrF** from the highly excited-state manifolds Kr***(6s, 4d), $F_2^*(a^3\Pi_u)$, $F_2^{**}(A^1\Pi_u)$ and so on could not be investigated in this model. These KrF** molecules then relaxed from the high- to the low-lying vibrational levels ($0 \leq v \leq 4$) of the collisionally mixed KrF*(B, C) state. It was also assumed that collisional mixing of the B and C states occurred rapidly compared with their vibrational relaxation processes. The main vibrational relaxation processes are the KrF**+Kr and KrF**+He ones [13]. On the other hand, highly excited KrF** and lasing KrF* species could be destroyed, mainly by electron-collision quenching (ECQ). The re-assessed rate constant for the ECQ process of 3×10^{-8} cm s^{-1} [17] was used, which value is close to that for the XeCl laser [24]. In comparison, rate-constant values for the other quenching processes such as KrF**/KrF* + F/F_2 were over an order of magnitude lower than that of the ECQ.

The computer algorithm [29] for solving the Boltzmann equation was obtained from the CPC Program Library [30].

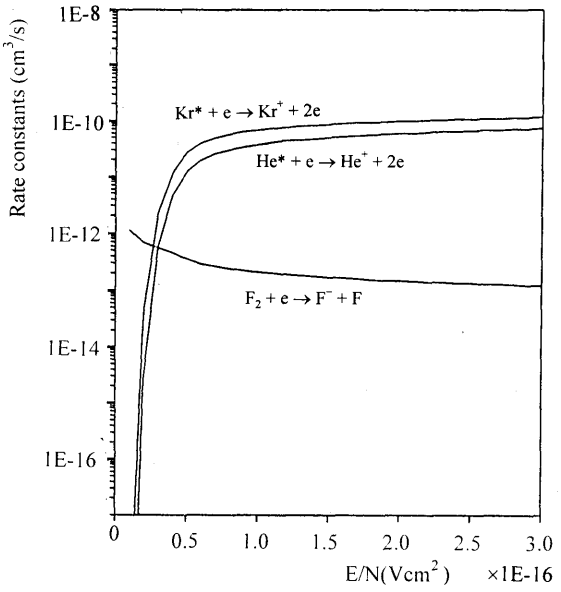
Table 1. There were 93 kinetic reactions among 30 different species: He, $He^*(2\,^3S$ and $2\,^1S)$, $He^{**}(2\,^1P)$, He^+, He^{2*}, He_2^+, Kr, $Kr^*(5s)$, $Kr^{**}(5p)$, $Kr^{***}(6s, 4d)$, Kr^{****}(higher), Kr^+, Kr^{2*}, Kr^{2+}, F_2, F, F^-, $F_2(v)$, $F_2^*(a\,^3\Pi_u)$, $F_2^{**}(A\,^1\Pi_u)$, $F_2^{***}(^1\Sigma_u^+)$, $F_2^{****}(H\,^1\Pi_u)$, F_2^+, KrF^*, KrF^{**}, Kr_2F^*, HeF^*, electron e, and photon $h\nu$.

Reaction number	Kinetics reaction	Rate constant	Reference
Electron–neutral species elastic collisions			
1	$e + He \rightarrow e + He$	Boltzmann equation (BE)	[31]
2	$e + Kr \rightarrow e + Kr$	BE	[32]
3	$e + F_2 \rightarrow e + F_2$	BE	[33]
Electron-impact excitation			
4	$e + Kr \rightarrow e + Kr^*(5s)$	BE	[33]
5	$e + Kr \rightarrow e + Kr^{**}(5p)$	BE	[33]
6	$e + Kr \rightarrow e + Kr^{***}(6s, 4d)$	BE	[33]
7	$e + Kr \rightarrow e + Kr^{****}$	BE	[33]
8	$e + He \rightarrow e + He^*$	BE	[34]
9	$e + He \rightarrow e + He^{**}(2\,^1S)$	BE	[34]
10	$e + He \rightarrow e + He^{**}(2\,^1P)$	BE	[34]
11	$e + Kr^* \rightarrow e + Kr^{**}(5p)$	6.4 (-7) cm^3 s^{-1}	[2]
Electron-impact ionization			
12	$e + Kr \rightarrow 2e + Kr^+$	BE	[35]
13	$e + Kr^* \rightarrow 2e + Kr^+$	BE	[36]
14	$e + Kr^{**} \rightarrow 2e + Kr^+$	1.8 (-7) cm^3 s^{-1}	[6]
15	$e + He \rightarrow 2e + He^+$	BE	[37]
16	$e + He^* \rightarrow 2e + He^+$	BE	[38]
17	$e + Kr_2^* \rightarrow 2e + Kr_2^+$	1.0 (-10) cm^3 s^{-1}	[22]
18	$e + He_2^* \rightarrow 2e + He_2^+$	1.0 (-10) cm^3 s^{-1}	[22]
Electron attachment			
19	$e + F_2 \rightarrow e + F_2(v)$	BE	[33]
20	$e + F_2 \rightarrow e + F_2^*(a^3\Pi_u)$	BE	[33]
21	$e + F_2 \rightarrow e + F_2^{**}(A^1\Pi_u)$	BE	[33]
22	$e + F_2 \rightarrow e + F_2^{***}(^1\Sigma_u^+)$	BE	[33]
23	$e + F_2 \rightarrow e + F_2^{*****}$	BE	[33]
24	$e + F_2 \rightarrow 2e + F_2^-$	BE	[33]
25	$e + F_2 \rightarrow F^- + F$	BE	[33]
26	$e + F \rightarrow F^-$	1.0 (-12) cm^3 s^{-1}	[6]
Electron quenching			
27	$e + Kr^{**} \rightarrow e + Kr^*$	8.0 (-7) cm^3 s^{-1}	[2]
28	$e + Kr^* \rightarrow e + Kr$	1.0 (-8) cm^3 s^{-1}	[20]
29	$e + He^* \rightarrow e + He$	4.2 (-9) cm^3 s^{-1}	[39]
30	$e + Kr_2{}^* \rightarrow e + 2Kr$	1.0 (-9) cm^3 s^{-1}	[22]
31	$e + He2* \rightarrow e + 2He$	1.0 (-10) cm^3 s^{-1}	[22]
32	$e + KrF^{**} \rightarrow e + Kr + F$	3.0 (-8) cm^3 s^{-1}	[17]
33	$e + KrF^* \rightarrow e + Kr + F$	3.0 (-8) cm^3 s^{-1}	[17]
34	$e + Kr_2F^* \rightarrow e + 2Kr + F$	1.0 (-7) cm^3 s^{-1}[16]	
Electron–ion recombination			
35	$e + Kr_2^+ \rightarrow Kr + Kr^*$	1.2 $(-6)(T_e/T_g)^{-0.5}$ cm^3 s^{-1}	[40]
36	$e + He_2^+ \rightarrow He^* + He + h\nu$‘	3.5 $(-10)(T_g/T_e)$ cm^3 s^{-1}	[39]
37	$e + He_2^+ \rightarrow He_2{}^* + h\nu$‘	1.0 $(-10)(T_g/T_e)$ cm^6 s^{-1}	[39]
38	$2e + He_2^+ \rightarrow He^* + He + e$	2.8 $(-20)(T_g/T_e)4$ cm^6 s^{-1}	[39]
39	$2e + He_2^+ \rightarrow He_2{}^* + e$	8.0 $(-21)(T_g/T_e)4$ cm^6 s^{-1}	[39]
40	$e + He_2^+ + He \rightarrow He_2{}^* + He + h\nu$‘	1.0 $(-27)(T_g/T_e)$ cm^6 s^{-1}	[39]
41	$2e + He^+ \rightarrow He + e$	1.0 $(-29)N_e(T_g/T_e)4.5$ cm^6 s^{-1}	[40]
42	$2e + He^+ \rightarrow He^* + e$	1.4 $(-27)N_eT_e$(eV) cm^6 s^{-1}	[40]
Penning ionization			
43	$Kr^* + Kr^* \rightarrow e + Kr^+ + Kr$	5.0 (-10) cm^3 s^{-1}	[22]
44	$Kr_2{}^* + Kr_2{}^* \rightarrow e + Kr_2^+ + 2Kr$	5.0 (-10) cm^3 s^{-1}	[22]
45	$He^* + Kr \rightarrow e + He + Kr^+$	2.2 (-10) cm^3 s^{-1}	[21]
46	$He^* + Kr + He \rightarrow e + Kr^+ + 2He$	8.6 (-31) cm^6 s^{-1}	[21]
47	$He_2{}^* + Kr \rightarrow e + Kr^+ + Kr$	1.5 (-10) cm^3 s^{-1}	[21]

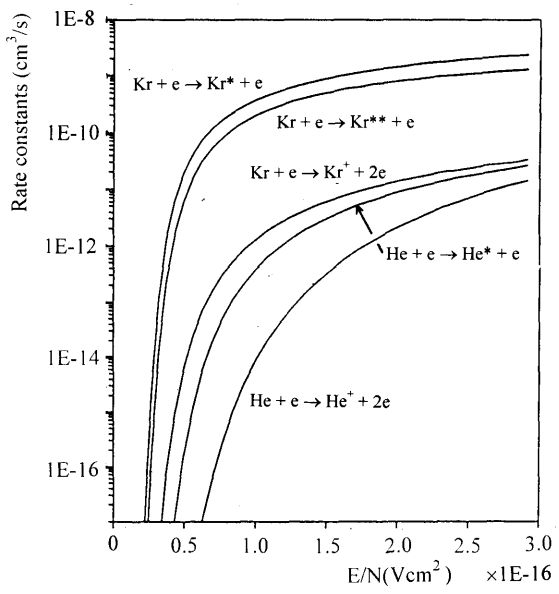
Table 1. (Continued)

Reaction number	Kinetics reaction	Rate constant	Reference
Excimer formation			
48	$Kr^* + F_2 \rightarrow KrF^{**} + F$	9.0 (−10) $cm^3\ s^{-1}$	[22]
49	$Kr^{**} + F_2 \rightarrow KrF^{**} + F$	8.1 (−10) $cm^3\ s^{-1}$	[6]
50	$Kr^+ + F^- \rightarrow KrF^{**}$	Function of [He]	[41]
51	$Kr_2^+ + F^- \rightarrow KrF^{**} + Kr$	Function of [He]	[41]
52	$Kr_2^* + F \rightarrow KrF^{**} + Kr$	3.0 (−10) $cm^3\ s^{-1}$	[22]
53	$Kr_2^* + F_2 \rightarrow KrF^{**} + Kr + F$	3.0 (−10) $cm^3\ s^{-1}$	[22]
54	$Kr_2^* + F_2 \rightarrow Kr_2F^* + F$	3.0 (−10) $cm^3\ s^{-1}$	[22]
55	$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$	2.6 (−32) $cm^6\ s^{-1}$	[22]
56	$Kr^* + Kr + He \rightarrow Kr_2^* + He$	2.7 (−32) $cm^6\ s^{-1}$	[22]
57	$He^* + 2He \rightarrow He_2^* + He$	2.0 (−34) $cm^6\ s^{-1}$	[22]
58	$He^+ + F^- \rightarrow HeF^*$		[22]
59	$He_2^+ + F^- \rightarrow HeF^* + He$		[22]
60	$KrF^{**} + Kr \rightarrow KrF^* + Kr$	1.0 (−10) $cm^3\ s^{-1}$	[13]
61	$KrF^{**} + He \rightarrow KrF^* + He$	5.0 (−11) $cm^3\ s^{-1}$	[13]
Excimer quenching			
62	$KrF^* + F_2 \rightarrow Kr + 3F$	4.5 (−10) $cm^3\ s^{-1}$	[42]
63	$KrF^* + Kr + He \rightarrow Kr_2F^* + He$	5.0 (−31) $cm^6\ s^{-1}$	[6]
64	$KrF^* + 2Kr \rightarrow Kr_2F^* + Kr$	6.7 (−31) $cm^6\ s^{-1}$	[15]
65	$KrF^* + 2He \rightarrow F + Kr + 2He$	5.0 (−31) $cm^6\ s^{-1}$	[20]
66	$KrF^* + Kr \rightarrow 2Kr + F$	2.0 (−12) $cm^3\ s^{-1}$	[43]
67	$KrF^* + F \rightarrow Kr + 2F$	7.8 (−10) $cm^3\ s^{-1}$	[22]
68	$KrF^{**} + F_2 \rightarrow Kr + 3F$	4.5 (−10) $cm^3\ s^{-1}$	[42]
69	$KrF^{**} + Kr + He \rightarrow Kr_2F^* + He$	5.0 (−31) $cm^6\ s^{-1}$	[20]
70	$KrF^{**} + 2Kr \rightarrow Kr_2F^* + Kr$	6.7 (−31) $cm^6\ s^{-1}$	[15]
71	$KrF^{**} + 2He \rightarrow F + Kr + 2He$	5.0 (−31) $cm^3\ s^{-1}$	[20]
72	$KrF^{**} + Kr \rightarrow 2Kr + F$	2.0 (−12) $cm^3\ s^{-1}$	[43]
73	$KrF^{**} + F \rightarrow Kr + 2F$	7.8 (−10) $cm^3\ s^{-1}$	[22]
74	$Kr_2F^* + F_2 \rightarrow 2Kr + F + F_2$	1.5 (−10) $cm^3\ s^{-1}$	[22]
75	$Kr_2F^* + 2Kr \rightarrow 4Kr + F$	2.9 (−31) $cm^6\ s^{-1}$	[40]
32	$KrF^{**} + e \rightarrow Kr + F + e$	3.0 (−8) $cm^3\ s^{-1}$	[16]
33	$KrF^* + e \rightarrow Kr + F + e$	3.0 (−8) $cm^3\ s^{-1}$	[16]
34	$Kr_2F^* + e \rightarrow 2Kr + F + e$	1.0 (−7) $cm^3\ s^{-1}$	[40]
Miscellaneous			
76	$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.5 (−31) $cm^6\ s^{-1}$	[44]
77	$Kr^+ + Kr + He \rightarrow Kr_2^+ + He$	1.9 (−31) $cm^6\ s^{-1}$	[21]
78	$He^+ + 2He \rightarrow He_2^+ + He$	1.0 (−31) $cm^6\ s^{-1}$	[22]
79	$He_2^+ + Kr \rightarrow Kr^+ + 2He$	1.8 (−11) $cm^3\ s^{-1}$	[22]
80	$He_2^+ + F_2 \rightarrow F^* + 2He + F$	1.0 (−9) $cm^3\ s^{-1}$	[21]
81	$F + F + He \rightarrow F_2 + He$	1.0 (−33) $cm^6\ s^{-1}$	[22]
Spontaneous emission			
82	$KrF^*/KrF^{**} \rightarrow Kr + F + h\nu$	1.5 (−8) s^{-1}	[45]
83	$Kr_2F^* \rightarrow 2Kr + F + h\nu'$	5.6 (−6) s^{-1}	[22]
84	$Kr_2^* \rightarrow 2Kr + h\nu''$	8.0 (−6) s^{-1}	[22]
85	$He_2^* \rightarrow 2He + h\nu'''$	3.6 (−6) s^{-1}	[22]
Stimulated emission			
86	$KrF^* + h\nu \rightarrow Kr + F + 2h\nu$	2.6 (−16) cm^2	[46]
Optical absorption			
87	$Kr^* + h\nu \rightarrow e + Kr^+$	1.3 (−18) cm^2	[22]
88	$Kr^{**} + h\nu \rightarrow e + Kr^+$	4.5 (−18) cm^2	[22]
89	$F^- + h\nu \rightarrow e + F$	5.5 (−18) cm^2	[21]
90	$F_2 + h\nu \rightarrow F + F$	1.6 (−20) cm^2	[47]
91	$Kr_2^+ + h\nu \rightarrow Kr^+ + Kr$	1.6 (−20) cm^2	[22]
92	$Kr_2F^* + h\nu \rightarrow 2Kr + F$	5.0 (−18) cm^2	[22]
93	$He^* + h\nu \rightarrow e + He^+$	4.8 (−18) cm^2	[21]

The electron superelastic collisions were taken into the EEDF calculation, but, owing to a lack of cross sectional data, the electron–electron collisions were excluded at this stage. Although their number densities were relatively



(*a*)



(*b*)

Figure 2. (*a*) Rate constants for the stepwise ionizations of Kr and He, and for electron attachment of F_2, as functions of E/N. (*b*) Rate constants for the electron-impact excitation and ionization of Kr and He as functions of E/N.

small, excited-state manifolds of Kr (Kr** and so on) and F_2 (F_2^*, F_2^{**} and so on) were included in the Boltzmann equation for the calculation of the electron energy distribution function. In particular, the percentage of F_2 in the laser gas mixture was above 1%, so it was necessary to include corresponding electron-impact vibrational, electronic and ionization processes in the discharge-excited kinetics. Owing to the large degeneracy of the Kr**(5p) state, this species could also be significantly populated [8] and also readily ionized by electrons and

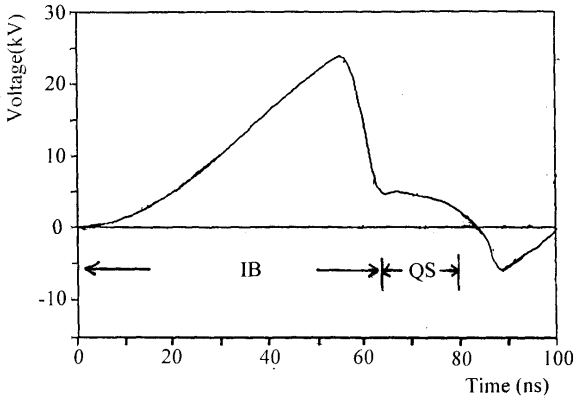


Figure 3. The computed voltage waveform for the laser discharge at 3 bar gas pressure and 22 kV charging voltage. IB, initial and beakdown phase; and QS, quasi-steady phase.

laser photons. Although the presence of Kr** species would increase the KrF formation and electron density, it decreased the F_2 density through the $Kr^+ + F_2$ and electron dissociation–attachment processes. Figure 2(*a*) shows the computed rate constants for stepwise ionization of the Kr and He metastable species, whereas those of the electron-impact excitation and ionization are presented in figure 2(*b*), each as a function of E/N.

The electric discharge in the laser channel was described by the following two equations:

$$L_1\,\mathrm{d}I_1/\mathrm{d}t + I_1R_1 + \int I_1/C_2\,\mathrm{d}t + \int I_1/C_1\,\mathrm{d}t - \int I_2/C_2\,\mathrm{d}t = V_0$$

$$L_2\,\mathrm{d}I_2/\mathrm{d}t + I_2R_2 + \int I_2/C_2\,\mathrm{d}t - \int I_1/C_2\,\mathrm{d}t = 0$$

where V_0 is the charging voltage; R_1 and R_2 the spark-gap and time-varying plasma resistances; L_1 and L_2, I_1 and I_2, C_1 and C_2 are, respectively, the inductances, currents and capacitances of the main and peaking circuits. The plasma resistance was given by $R_2 = d/(eA\mu_e n_e)$, where d is the laser-channel gap, e the electronic charge, A the discharge cross sectional area, μ_e the electron mobility and n_e the electron density. The two parameters, μ_e and n_e, which usually greatly determine the temporal behaviour of the plasma resistance, were computed from the Boltzmann equation at each computational time step. The initial value for the plasma resistance R_2 was set to several tens of mega-ohms. The discharge currents I_1 and I_2 were iterated until $\Delta I/I \leq 0.000\,01$ was achieved. An E/N value was calculated from $E/N = I_2R_2/(Nd)$; where E is the reduced electric field across the laser channel and N is the total particle number density.

4. The temporal evolution of species

The initial densities of all the discharge species were set to be 10^2 cm^{-3}, which did not have much effect on the model predictions of the laser output energy and the discharge

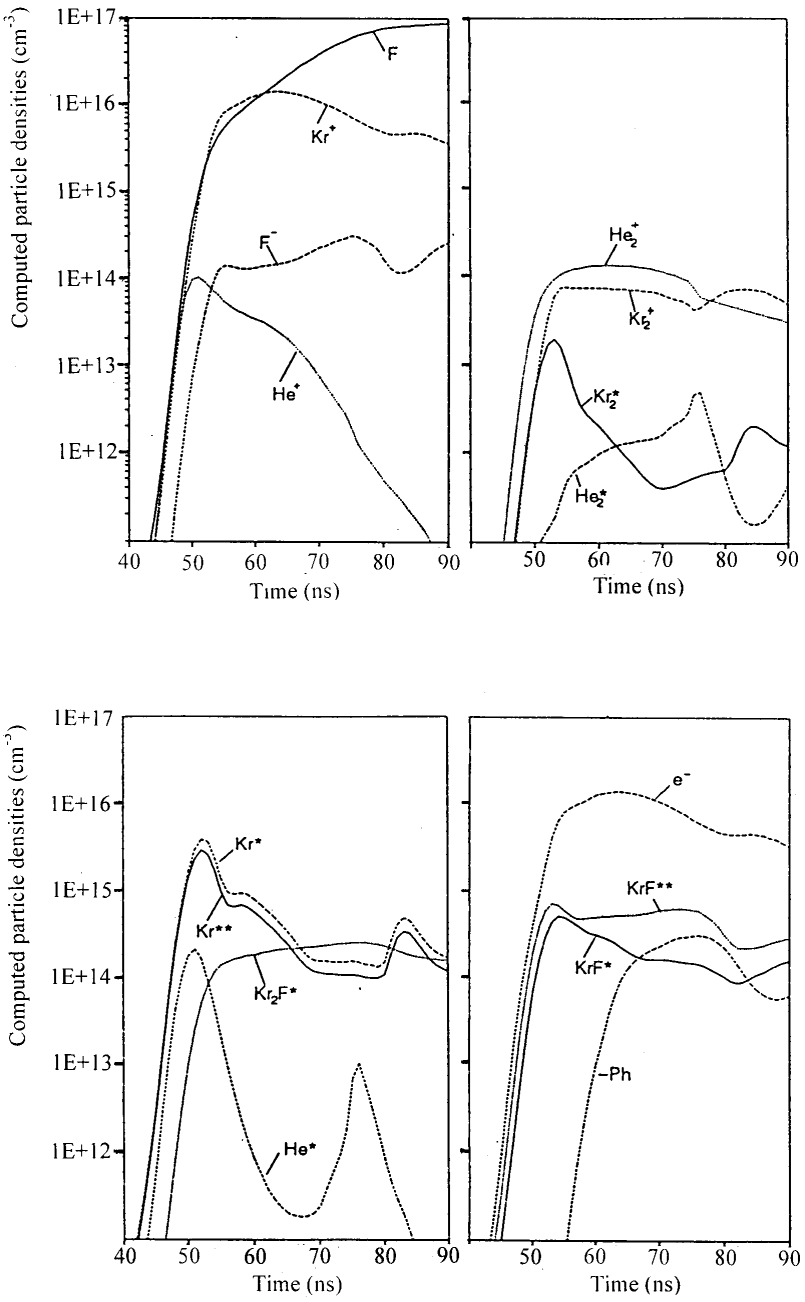


Figure 4. The temporal evolution of 16 selected discharge-excited species: F, Kr^+, F^-, He^+, He_2^+, Kr_2^+, Kr_2^*, He_2^*, Kr^*, Kr^{**}, Kr_2F^*, He^*, e^-, KrF^{**}, KrF^* and $Ph(h\nu)$.

characteristics. Although UV-pre-ionization increased the Kr^*, Kr^{**}, Kr^+ and electron densities to about 10^7 cm^{-3}, these were not found to affect the subsequent discharge and laser output significantly. The laser discharge could be broadly divided into (i) the initial and breakdown phase, and (ii) the quasi-steady phase, as indicated in the computed voltage waveform in figure 3.

The main kinetics during the initial and breakdown phase was associated with electron production at the low pulsed voltage. During this phase, the electron loss was faster than the production by the two main processes of electron-impact and Penning ionizations, $e^- + Kr$ and $Kr + He^*$ respectively. As the pulsed voltage was increased further, electron-impact ionization of He, $e^- + He$, became an additional electron source. The relative importance of these electron production channels could be inferred from their rate coefficients in figures 2(a) and 2(b). When the electron density was increased to about 10^{12} cm^{-3}, the applied voltage across the laser channel began to drop rapidly within 10 ns, indicating an electrical breakdown. The plasma resistance R_2 of the laser discharge decreased inversely with respect to the electron density, whereas the electron mobility μ_e did not increase significantly. After this, the laser discharge entered a quasi-steady phase, as

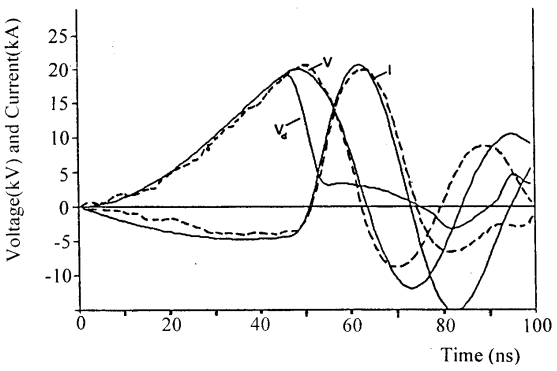


Figure 5. Comparisons between the computed (full line) and measured (dotted) discharge waveforms. V, voltage; I, current; and V_d, the computed laser-channel voltage.

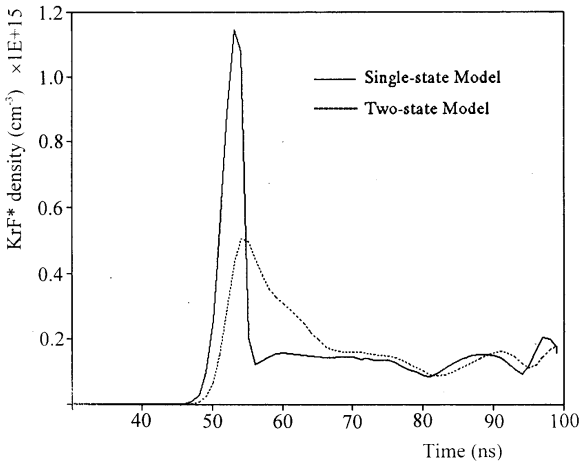


Figure 6. A comparison between the KrF^* densities using the single-state and two-state models for the upper laser level.

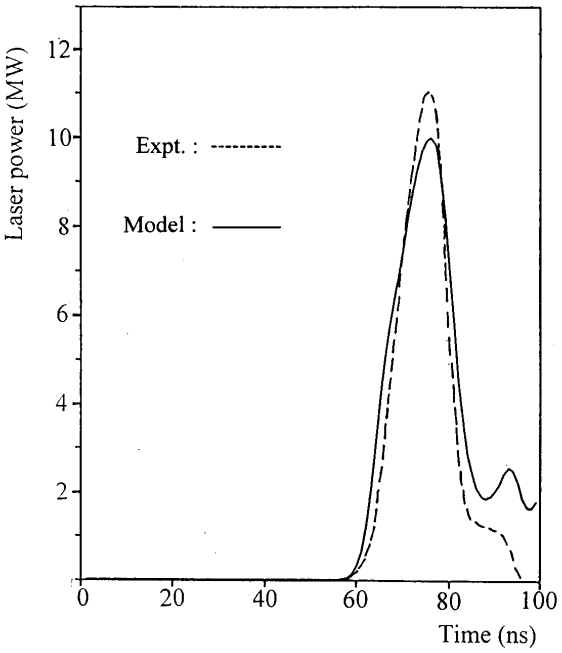


Figure 7. A comparison between the computed and measured laser pulse characteristics.

reflected by a voltage plateau.

The plots in figures 4 show the temporal evolution of 16 selected discharge species whose densities were at least three orders of magnitude higher than the rest. It was computed that, during the quasi-steady phase, the respective densities for excited-state manifolds of Kr^* and Kr^{**} decreased by more than an order of magnitude while that of the vibrationally excited KrF^{**} species remained fairly steady. Figures 5(*a*) and (*b*) show the time-dependences of the formation and loss of the KrF^{**} and KrF^* species. The $Kr_2{}^*$ species was found to contribute to the photon-absorbing species Kr_2F^*. The computed results showed that ECQ of the KrF^*, KrF^{**} and Kr_2F^* species contributed to the rising atomic density of F. On the other hand, ECQ reduced the He^* metastable species density. The density of He_2^* was increased mainly the electron–ion recombination excitation, $e^- + He_2^+$.

The rates of formation of excited KrF molecules were compared for two separate cases of (i) one-state and (ii) two-state upper laser levels. The second case has been explained before; the first case assumed that excited KrF molecules were formed immediately at the low-lying vibrational levels (KrF^*) which could be directly extracted for the laser oscillation. There was a rapid build-up of the KrF^* species for the first case compared with that for the second. As a result the peak density of KrF^* was about 2.5 times higher in the first case, which had a significant effect on the laser output. The predicted laser output energy for the first case was 5–6 times higher than that for the second case. By assuming a single-state upper laser level, the present kinetics model predicted an earlier laser emission, longer pulse duration and higher output energy compared with the experimental results.

5. Comparison

In figure 6, the computed and measured voltage and current are compared, showing reasonably good agreement until their reversed cycles. The main reason for these discrepancies may be explained in terms of the discharge instabilities because numerous filaments were usually observed above the wire-mesh electrode. These could reduce the plasma resistance towards the end of the quasi-steady phase, which resulted in a larger reversal swing in the discharge waveforms.

The predicted and measured laser pulse shape are shown in figure 7. In comparison, the computed pulse width at FMHW was about 15 ns whereas the measured value was 12–13 ns, which showed a reasonably good agreement. The validity of the two-state upper laser level for the present model may also be supported by the onset time of the laser emission, which was accurately predicted to occur at about 60 ns. When the gas pressure and the charging voltage were increased, this onset time was shortened accordingly both in experiment and in the model prediction. In contrast,

by assuming a single-state upper laser level, this kinetic model would consistently predict an early onset time, a longer laser pulse and higher output energy. The present model, however, usually predicted a tail section in the laser pulse which was slightly longer than that observed in the photodiode signal. A shorter tail section in the experiment may or may not support the above suggestion of discharge instabilities towards the end of the quasi-steady phase, which terminate the laser oscillation.

Figures 8(*a*) and (*b*) compare the predicted and measured output energies as a function of gas pressure and charging voltage respectively, showing agreements of within 5%. The predicted laser output energy, however, is usually higher than the measured value. This may be explained in terms of the vibrational relaxation $KrF^{**} + He$, for which the rate constant 5×10^{-11} cm^3 s^{-1} was an upper limit. The other vibrational relaxation process $KrF^{**}+Kr$, although with twice as large a rate constant of 1×10^{-10} cm^3 s^{-1}, would contribute less to the KrF^* species owing to the relatively low Kr pressure. At the higher charging voltage of $V_0 = 24$ kV, it was observed that the discrepancy between the predicted and measured laser output energy was increased. This could be due to an increasing electron–electron collision rate, which was not taken into consideration by the Boltzmann equation, owing to a lack of cross section data.

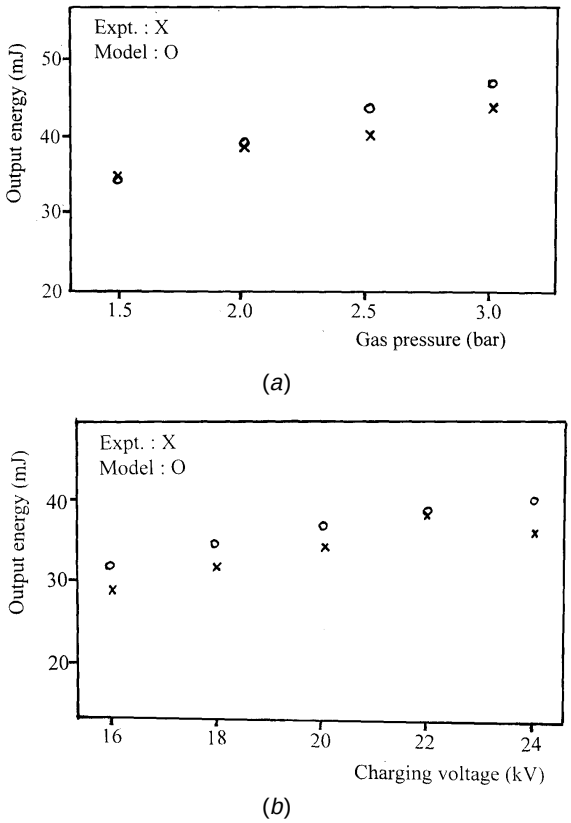


Figure 8. A comparison between the computed and measured laser output energies for (*a*) different gas pressures and (*b*) different charging voltages.

6. Discussion and conclusion

The reasonably good agreement between the predicted and measured results on the laser discharge and output characteristics may suggest that the basic kinetics which are essential for a better understanding of an avalanche discharge-pumped KrF laser had been considered in this kinetic model. The laser output energy as predicted by the model could agree to within 5% with the measured value. The discrepancies in the reversed cycles of the discharge waveforms were largely ascribed to the possible discharge instabilities after the quasi-steady phase, which may explain also why a predicted tail section in the laser pulse was not observed in the photodiode signal. The tail section may largely explain why a higher laser output energy is consistently predicted by this kinetic model.

Although the assumption of the two-state upper laser level produced the bulk of the agreement between the predicted and measured laser output energy, the immediate formation of KrF^{**} and no KrF^* species may not be entirely correct. This may partially explain the predicted pulse width which was about 15% more than the measured value, owing to a prolonged vibrational relaxation. The doubly excited Kr^{**} species was found to be a significant source for the formation of KrF^{**}, which had not been considered in the early models. In addition, this model included also a number of kinetic processes involving the F_2 species when its gas content exceeded 0.01%, which was found to influence the EEDF. The contribution from the excited-state manifolds of F_2 and from those above Kr^{**} to the KrF^{**} formation could not be calculated in this model, owing to a lack of relevant cross section data. Although these species were comparatively small in number densities, they were included in the Boltzmann equation. The electron–electron collisions, which were expected to increase the computational time and affect the discharge kinetics at high charging voltages, were not considered simply because their cross section data were not available. Despite the lack of these fine details which could not be considered at this stage, the present model may represent a much improved version compared to the early ones.

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